

```
erfan@erfan:~$ mkdir ~/monitoring-project && cd ~/monitoring-project
```

```
erfan@erfan:~/monitoring-project$ mkdir -p nginx/html nginx/logs prometheus grafana/provisioning/datasources grafana/dashboards scripts
```

```
erfan@erfan:~/monitoring-project$ tree
```

```
Command 'tree' not found, but can be installed with:
```

```
sudo snap install tree # version 2.1.3+pkg-5852, or
```

```
sudo apt install tree # version 2.0.2-1
```

```
See 'snap info tree' for additional versions.
```

```
erfan@erfan:~/monitoring-project$ ^C
```

```
erfan@erfan:~/monitoring-project$ sudo snap install tree
```

```
[sudo] password for erfan:
```

```
tree 2.1.3+pkg-5852 from 林博仁 Buo-ren Lin (brlin) installed
```

```
erfan@erfan:~/monitoring-project$ tree
```

```
locales-launch: Data of en_US locale not found, generating, please wait...
```

```
.
├── grafana
│   ├── dashboards
│   ├── provisioning
│   │   └── datasources
├── nginx
│   ├── html
│   └── logs
├── prometheus
└── scripts
```

```
10 directories, 0 files
```

```
erfan@erfan:~/monitoring-project$
```

```
cat > nginx/html/index.html << 'EOF'
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>DevOps Monitoring Demo</title>
```

```
  <style>
```

```
    body { font-family: Arial; text-align: center; margin-top: 100px; background: #f0f0f0; }
```

```
    h1 { color: #2c3e50; }
```

```
  </style>
```

```
</head>
```

```
<body>
```

```
  <h1>Hello from Nginx! 🚀</h1>
```

```
  <p>است Junior DevOps پروژه Monitoring این صفحه برای تست.</p>
```

```
  <p>افزایش پیدا میکنه Grafana کن و ببین درخواستها در Refresh</p>
```

```
</body>
```

```
</html>
```

```
EOF
```

```
GNU nano 6.2 nginx.conf
cat > nginx/nginx.conf << 'EOF'
events {}

http {
    server {
        listen 80;
        server_name localhost;

        location / {
            root /usr/share/nginx/html;
            index index.html;
        }

        location /stub_status {
            stub_status on;
            access_log off;
        }

        access_log /var/log/nginx/access.log;
        error_log /var/log/nginx/error.log;
    }
}
EOF
```

[Read 23 lines]

```
cat > prometheus/prometheus.yml << 'EOF'
```

```
global:
```

```
  scrape_interval: 15s
```

```
scrape_configs:
```

```
- job_name: 'prometheus'
```

```
  static_configs:
```

```
    - targets: ['localhost:9090']
```

```
- job_name: 'cadvisor'
```

```
  static_configs:
```

```
    - targets: ['cadvisor:8080']
```

```
- job_name: 'nginx'
```

```
  static_configs:
```

```
    - targets: ['nginx-exporter:9113']
```

```
EOF
```

```
apiVersion: 1
```

```
datasources:
```

```
- name: Prometheus
  type: prometheus
  access: proxy
  url: http://prometheus:9090
  isDefault: true
```

```
GNU nano 6.2 docker-compose.yml *
cat[]> docker-compose.yml << 'EOF'
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: nginx
    ports:
      - "80:80"
    volumes:
      - ./nginx/html:/usr/share/nginx/html:ro
      - ./nginx/nginx.conf:/etc/nginx/nginx.conf:ro
      - ./nginx/logs:/var/log/nginx
    networks:
      - monitoring
    restart: unless-stopped

  cadvisor:
    image: gcr.io/cadvisor/cadvisor:latest
    container_name: cadvisor
    ports:
      - "8080:8080"
    volumes:
      - /:/rootfs:ro
      - /var/run:/var/run:ro
      - /sys:/sys:ro
      - /var/lib/docker:/var/lib/docker:ro
      - /dev/disk/:/dev/disk:ro
    networks:
      - monitoring
    restart: unless-stopped
    privileged: true

  nginx-exporter:
    image: nginxinc/nginx-prometheus-exporter:latest
    container_name: nginx-exporter
    ports:
      - "9113:9113"
    command:
```

```
erfan@erfan:~/monitoring-project$ nano index.html
erfan@erfan:~/monitoring-project$ nano nginx.conf
erfan@erfan:~/monitoring-project$ nano nginx.conf
erfan@erfan:~/monitoring-project$ nano prometheus.yml
erfan@erfan:~/monitoring-project$ nano nano grafana/provisioning/datasources/prometheus.yml
erfan@erfan:~/monitoring-project$ nano grafana/provisioning/datasources/prometheus.yml
erfan@erfan:~/monitoring-project$ nano docker-compose.yml
erfan@erfan:~/monitoring-project$ nano docker-compose.yml
erfan@erfan:~/monitoring-project$ ls -r
scripts prometheus.yml prometheus nginx.conf nginx index.html grafana docker-compose.yml
erfan@erfan:~/monitoring-project$ docker compose up -d
```

```
erfan@erfan:~/monitoring-project$ docker compose up -d
```

```
WARN[0000] /home/erfan/monitoring-project/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

```
[+] up 32/47
```

```
.: Image ghcr.io/google/cadvisor:latest [#####] 19.31MB / 30.66MB Pulling 22.6s
.: Image nginx/nginx-prometheus-exporter:latest [#####] 5.231MB / 5.231MB Pulling 22.6s
.: Image prom/prometheus:latest [#####] 15.73MB / 117MB Pulling 22.6s
.: Image grafana/grafana:latest [#####] 32.48MB / 351.9MB Pulling 22.6s
.: Image nginx:alpine [#####] 14.76MB / 28.55MB Pulling 22.6s
```



```
erfan@erfan:~/monitoring-project$ docker compose down
```

```
WARN[0000] /home/erfan/monitoring-project/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

```
[+] down 6/6
```

```
✓ Container grafana          Removed 0.0s
✓ Container prometheus       Removed 0.0s
✓ Container cadvisor         Removed 2.4s
✓ Container nginx-exporter   Removed 2.3s
✓ Container nginx            Removed 0.8s
✓ Network monitoring-project_monitoring Removed 0.2s
```

```
erfan@erfan:~/monitoring-project$
```

```
erfan@erfan:~/monitoring-project$ docker compose up -d
```

```
WARN[0000] /home/erfan/monitoring-project/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

```
[+] up 6/6
```

```
✓ Network monitoring-project_monitoring Created 0.3s
✓ Container cadvisor          Created 0.7s
✓ Container nginx             Created 0.8s
✓ Container nginx-exporter    Created 0.2s
✓ Container prometheus        Created 0.4s
✓ Container grafana           Created 0.4s
```

```
erfan@erfan:~/monitoring-project$ docker compose ps
```

```
WARN[0000] /home/erfan/monitoring-project/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
cadvisor	ghcr.io/google/cadvisor:latest	"/usr/bin/entrypoint..."	cadvisor	26 seconds ago	Up 23 seconds (health: starting)	0.0.0.0:8080->8080/tcp
grafana	grafana/grafana:latest	"/run.sh"	grafana	25 seconds ago	Up 9 seconds	0.0.0.0:3000->3000/tcp
nginx	nginx:alpine	"/docker-entrypoint..."	nginx	26 seconds ago	Up 23 seconds	0.0.0.0:80->80/tcp
nginx-exporter	nginx/nginx-prometheus-exporter:latest	"/usr/bin/nginx-prom..."	nginx-exporter	25 seconds ago	Up 21 seconds	0.0.0.0:9113->9113/tcp
prometheus	prom/prometheus:latest	"/bin/prometheus --c..."	prometheus	25 seconds ago	Up 19 seconds	0.0.0.0:9090->9090/tcp

```
erfan@erfan:~/monitoring-project$ ^C
```

```
erfan@erfan:~/monitoring-project$
```

Hello from Nginx! 🚀

This is a demo page for Junior DevOps Monitoring Project.

Refresh this page several times and watch the requests increase in Grafana!

Project Status: Running Successfully



Welcome to Grafana

Email or username

Password

Log in

[Forgot your password?](#)

SEARCH...

MySQL

+

do we run mysql?

I'll help you determine if you're running MySQL in your Grafana environment. To answer this question accurately, I need to check if there are any MySQL datasources configured in your Grafana instance.

✓

list available datasources

▼

Let me search for any dashboards that might be related to MySQL, as this could indicate MySQL usage in your environment:

🔍

search for dashboards

▼

Based on my checks, I don't see any evidence that you're running MySQL in your Grafana environment:

1. There are no MySQL datasources configured in your Grafana instance
2. There are no dashboards with "mysql" in their titles.

The only datasource I found is the default Grafana internal datasource, which is not MySQL.

If you believe MySQL should be configured or if you're looking to set up MySQL as a datasource, I can provide guidance on how to do that. Would you like information about adding MySQL as a datasource to your Grafana instance?

📄

demo-logs

📄

demo-metrics

Ask questions, go places, make changes, anything.

➡

Send

🗉

Give feedback

98%

NEW

Grafana Assistant is now available to OSS users

🌟

Use AI to handle everything from config to query generation

- Explore your data by prompting Assistant to generate queries for you
- Create comprehensive dashboards in minutes
- Onboard new team members in days, not weeks

<

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>

1/4

🔗

Show me



/

root

- Docker Containers
- Podman Containers
- Subcontainers

/init.scope

/system.slice

/user.slice

Welcome to Connections



Add new connection

Connect to a new data source

[View configured data sources](#)

Manage your existing data source connections

+ Add new data source

DevOps Monitoring Demo

Prometheus Time Series Collection ar

Prometheus - Data sources - Cor x

cAdvisor - /

+

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🔖

⚠️ Not secure 192.168.2.142:3000/connections/datasources/edit/PBFA97CFB590B2093

🔗 ↗️ 🔒 🚧

🛠️ 📄 | 📄 📄 ⭐️ 🛡️ ☰

🪴

📺 BTCUSDT

🤖 ChatGPT

🔗 grok.com

🌐 DeepSeek

📖 NotebookLM

🧠 Perplexity

🔄 LibreMobileOS Chat...

🎓 ورود به کلاس دانشگاه

🏠 ارزنگ

🎓 آموزشیار

📧 حامی

💡 نسخه وب - iGap

>>

☰ ⚙️ Connections > Data sources > Prometheus

🔍 Search...

ctrl+k

+ ▾

?

🐼

Custom query parameters ⓘ

Example: max_source_resolution=5m&tin

HTTP method ⓘ

POST ▾

Series limit ⓘ

40000

Use series endpoint ⓘ

☐

Exemplars

No exemplars configurations

✓ Successfully queried the Prometheus API.

Next, you can start to visualize data by [building a dashboard from scratch](#) or by querying data in the [Explore view](#).

Open in Metrics Drilldown

Delete

Test

DevOps Monitoring Demo

Prometheus Time Series Collection ar

Complete DevOps Monitoring St x

cAdvisor - /

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🔖

⚠️ Not secure 192.168.2.142:3000/d/adkfk8c/complete-devops-monitoring-stack-enhanced?orgId=...

🔍

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BTCUSDT

ChatGPT

grok.com

DeepSeek

NotebookLM

Perplexity

LibreMobileOS Chat...

ورود به کلاس دانشگاه

ارژنگ

آموزشیار

حامی

نسخه وب - iGap

>>

☰

⚙️ Dashboards > Complete DevOps Monitoring Stack (Enhanced) ☆

🔍 Search... ctrl+k

+ ▾

❓

👤

job

All ▾

+

⏪

🕒 Last 5 minutes ▾

⏩

🔍

🔄 Refresh

5s ▾

🔗 Share ▾

💾 Save ▾

✂️ Exit edit

Running Containers

48

Nginx Up

1

Active Connections

1

Request Rate (req/s)

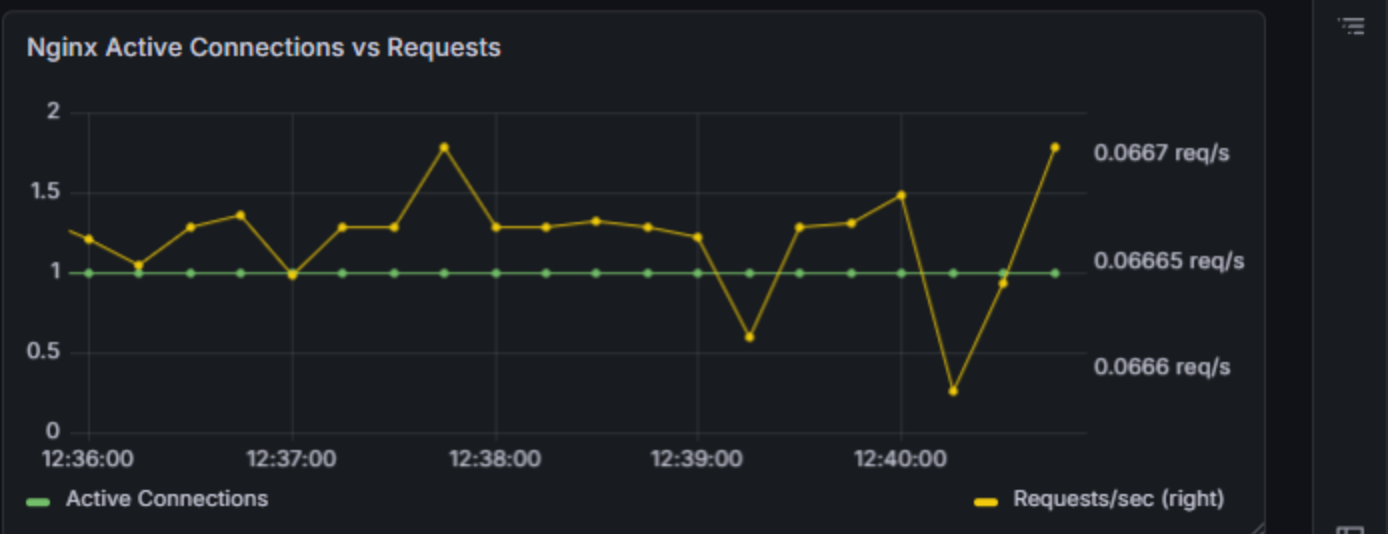
0.0667 req/s

Nginx Connections (by state)



State	Value
Reading	1
Writing	1
Waiting	1

Nginx Active Connections vs Requests



Time	Active Connections	Requests/sec (right)
12:36:00	1	1.2
12:37:00	1	1.0
12:38:00	1	1.8
12:39:00	1	1.2
12:40:00	1	1.5

CPU Usage per Container (top 5)

Prometheus Targets Scrape Status

DevOps Monitoring Demo

Prometheus Time Series Collection ar

Complete DevOps Monitoring St x

cAdvisor - /

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🔖

⚠️ Not secure 192.168.2.142:3000/d/adkfk8c/complete-devops-monitoring-stack-enhanced?orgId=...

🔍

🔗

🛡️

🚩

☰

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📺 BTCUSDT

🤖 ChatGPT

🔗 grok.com

🌐 DeepSeek

📝 NotebookLM

🧠 Perplexity

🔄 LibreMobileOS Chat...

🎓 ورود به کلاس دانشگاه

🎯 ارزنگ

🎓 آموزشیار

👤 حامی

🟢 نسخه وب - iGap

➡

☰ ⚙️

Dashboards > Complete DevOps Monitoring Stack (Enhanced) ☆

🔍 Search... ctrl+k

+ ▾

❓

🐙

job All ▾ +

⏪ ⌚ Last 5 minutes ▾ ⏩ 🔍

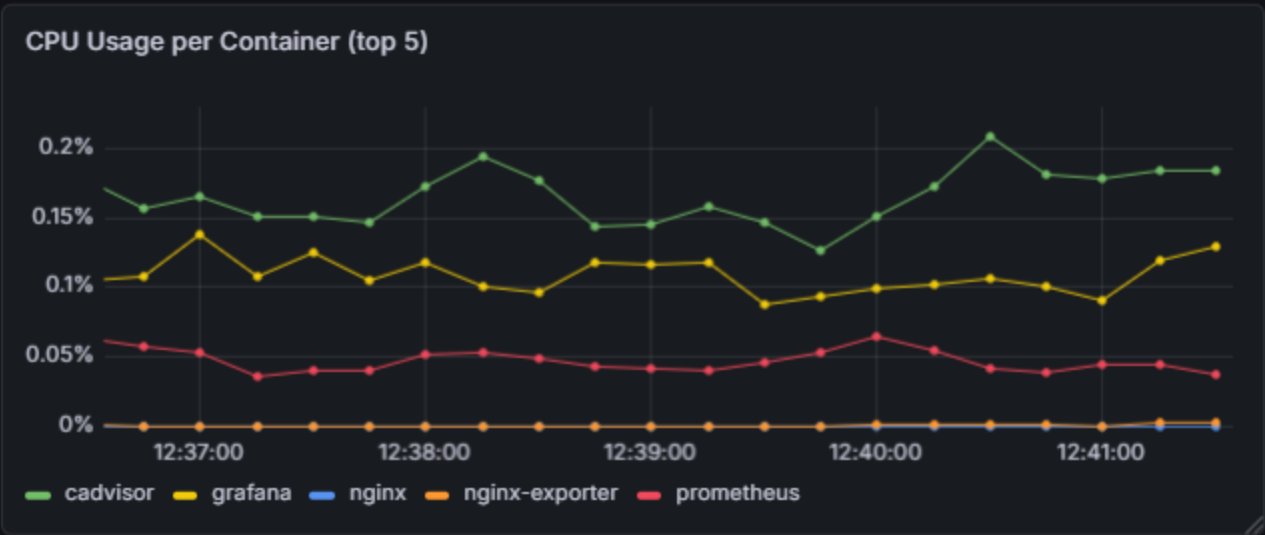
🔄 Refresh 5s ▾

🔗 Share ▾

💾 Save ▾

🗑️ Exit edit

CPU Usage per Container (top 5)



This line chart displays the CPU usage percentage for the top 5 containers over a 5-minute period. The y-axis ranges from 0% to 0.2% in 0.05% increments. The x-axis shows timestamps from 12:37:00 to 12:41:00. The legend identifies five series: cadvisor (green), grafana (yellow), nginx (blue), nginx-exporter (orange), and prometheus (red). cadvisor shows the highest usage, peaking at approximately 0.2% around 12:40:30. grafana follows, with usage fluctuating between 0.08% and 0.15%. prometheus usage is around 0.04-0.06%. nginx and nginx-exporter maintain very low, near-zero CPU usage throughout the period.

Container	Usage (approx. %)
cadvisor	0.15 - 0.20
grafana	0.08 - 0.15
nginx	0.00 - 0.01
nginx-exporter	0.00 - 0.01
prometheus	0.04 - 0.06

Prometheus Targets Scrape Status

job	instance	__name__
cadvisor	cadvisor:8080	up
prometheus	localhost:9090	up
nginx	nginx-exporter:9113	up

Memory Usage per Container (top 5)



This line chart displays the memory usage in bytes for the top 5 containers over a 5-minute period. The y-axis ranges from 0 B to 1 GiB, with major ticks at 0 B, 256 MiB, 512 MiB, 768 MiB, and 1 GiB. The x-axis shows timestamps from 12:37:00 to 12:41:00. The legend identifies five series: cadvisor (green), grafana (yellow), nginx (blue), nginx-exporter (orange), and prometheus (red). grafana shows the highest memory usage, starting around 768 MiB and increasing to nearly 1 GiB by the end of the period. cadvisor and prometheus maintain low memory usage, around 128 MiB. nginx and nginx-exporter use very little memory, near 0 B.

Container	Usage (approx. MiB)
grafana	768 - 1024
cadvisor	128
prometheus	128
nginx	0 - 16
nginx-exporter	0 - 16

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